

STROBE checklist for the cohort study

Landmark-specific transcriptomic and proteomic associations with 60-day mortality in Day-1-defined sepsis-induced coagulopathy

Locations refer to manuscript sections and numbered supplementary tables. Items not fully reported are identified explicitly; these entries do not imply that additional analyses were performed.

Topic	Item	Recommendation	Current reporting location or qualification
Title and abstract	1 (a)	(a) Indicate the study's design with a commonly used term in the title or the abstract	Title and Abstract: longitudinal cohort design is identified.
Title and abstract	1 (b)	(b) Provide in the abstract an informative and balanced summary of what was done and what was found	Abstract: cohort, landmark-specific sample and event counts, principal findings and interpretation limits.
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported	Introduction: SIC biology, molecular heterogeneity and rationale for landmark analyses.
Objectives	3	State specific objectives, including any prespecified hypotheses	Introduction, final paragraph: study objectives. Exploratory analyses are distinguished in Methods and Supplementary Methods; prospectively documented selection is not claimed for every display.
Study design	4	Present key elements of study design early in the paper	Methods: Study design and data source; Analytic framework and estimands.
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection	Methods: Study design and data source; Participants, SIC definition and outcome; Longitudinal sample linkage and time-specific risk sets. Recruitment period and 60-day horizon are reported.
Participants	6 (a)	(a) Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up	Methods: Participants, SIC definition and outcome; Longitudinal sample linkage and time-specific risk sets. Results: Cohort construction and longitudinal molecular availability. Figure 1.
Participants	6 (b)	(b) For matched studies, give matching criteria and number of exposed and unexposed	Not applicable: this is not a matched cohort study.
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable	Methods: Participants, SIC definition and outcome; Clinical descriptive and exploratory survival analyses; molecular and cross-omic model sections. Supplementary Table S1.

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Data sources/ measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group	Methods: Molecular data generation in the parent cohort; RNA-seq processing; Plasma protein processing and quality control. Supplementary Methods and Table S1. Protein matrix scale and experimental QC grouping remain incompletely documented.
Bias	9	Describe any efforts to address potential sources of bias	Methods: delayed entry, centre stratification, availability weighting and sensitivity analyses. Discussion: selection, clinical confounding, cell composition and measurement limitations. Tables S10–S18.
Study size	10	Explain how the study size was arrived at	Methods: all eligible patients and available molecular measurements were included; no prospective sample-size calculation.
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why	Methods: Clinical descriptive and exploratory survival analyses; RNA/protein standardisation; pathway scoring and availability models. Tables S1–S2 define contrasts and units.
Statistical methods	12 (a)	(a) Describe all statistical methods, including those used to control for confounding	Methods: clinical, molecular, cross-omic and IPW statistical specifications. Supplementary Methods: implementation, FDR families, diagnostics and sensitivity comparisons.
Statistical methods	12 (b)	(b) Describe any methods used to examine subgroups and interactions	No primary subgroup or interaction analysis is reported. Landmark-specific analyses and the fixed cross-omic pathway panel are described separately, not as formal interaction tests.
Statistical methods	12 (c)	(c) Explain how missing data were addressed	Methods: Missing data, multiplicity and interpretation hierarchy. Supplementary Methods: RNA summation and source-missingness handling; protein completeness and QC; availability weighting. Tables S1, S8 and S9.
Statistical methods	12 (d)	(d) If applicable, explain how loss to follow-up was addressed	Methods: Participants, SIC definition and outcome; follow-up is administratively censored at 60 days. Later assay non-observation is treated separately from mortality follow-up in Day-5 availability methods.
Statistical methods	12 (e)	(e) Describe any sensitivity analyses	Methods and Supplementary Methods: PH-restricted, protein-intensity, severity-adjusted, centre-omission, IPW, matched cross-omic and protein CV comparisons. Tables S3–S8 and S10–S18.
Participants	13* (a)	(a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed	Results: Cohort construction and longitudinal molecular availability; Figure 1. Tables S8, S12 and S14 provide analysis-population details.
Participants	13* (b)	(b) Give reasons for non-participation at each stage	Results: structural non-availability and zero-observation centre exclusions are distinct. Figure 1; Tables S8 and S14.

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Participants	13* (c)	(c) Consider use of a flow diagram	Figure 1 presents cohort linkage, molecular risk sets and Day-5 availability.
Descriptive data	14* (a)	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders	Table 1 and Supplementary Table S9: baseline characteristics by 60-day outcome. Tables S12 and S14: molecular availability and excluded-survivor descriptive comparisons.
Descriptive data	14* (b)	(b) Indicate number of participants with missing data for each variable of interest	Supplementary Tables S1, S8 and S9: variable definitions and missingness. Table S14: missing counts for excluded-survivor baseline summaries. RNA source-missingness handling is described in Supplementary Methods.
Descriptive data	14* (c)	(c) Summarise follow-up time (eg, average and total amount)	The 60-day administrative horizon and landmark eligibility are reported in Methods. Mean/median observed follow-up and total person-time are not separately summarised; the horizon should not be interpreted as those summaries.
Outcome data	15*	Report numbers of outcome events or summary measures over time	Results: 84 deaths in the 504-patient cohort; RNA events 84/67/53 and protein events 27/18/14 at Days 1/3/5. Abstract and Tables S3/S5.
Main results	16 (a)	(a) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included	Results and Figures 2–4; complete estimates and uncertainty in Tables S2–S8. Primary molecular models are centre-stratified, not severity-adjusted; additional adjustment is identified in Tables S10/S15.
Main results	16 (b)	(b) Report category boundaries when continuous variables were categorized	Continuous clinical variables use stated increments rather than arbitrary outcome-dependent categories. SIC eligibility and infection-source categories are defined in Methods and Tables S1–S2.
Main results	16 (c)	(c) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period	Not performed: this study estimates mortality associations and does not develop or validate an absolute-risk prediction model.
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses	Exploratory and sensitivity results are reported in Results, Supplementary Methods and Tables S10–S18, alongside the primary and PH/intensity/IPW outputs in S3–S8.
Key results	18	Summarise key results with reference to study objectives	Discussion, opening paragraphs: main findings in relation to the landmark and cross-compartment objectives.
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias	Discussion, Strengths and limitations: selection, residual confounding, cell composition, small protein event counts, missing-source semantics and QC uncertainty. Supplementary Methods and Table S18 define the protein CV sensitivity boundaries.

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Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence	Discussion and Conclusions: non-causal mortality associations, multiplicity, prior CMAISE work, compartment differences and constraints on SIC specificity, trajectories and prediction.
Generalisability	21	Discuss the generalisability (external validity) of the study results	Discussion: multicentre setting, selected protein subset, centre/sample availability and need for independent validation; no claim of external validation from centre-omission analyses.
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based	Funding statement: current-study support and funder role. Parent-cohort source publications are cited; their full funding details are not reproduced in this manuscript.

* Where applicable, descriptive and outcome information is reported separately for outcome groups or the stated comparison populations. Recommendations retained from the original submitted STROBE cohort checklist.